Travelling Waves

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Travelling Waves

INTRODUCTION

Till chapter Rotational Motion (*Rigid Body Dynamics*) we had analysed mechanics of rigid body, now in this chapter we will study the effect of forces on non-rigid body that is elastic body.

On application of force, distances between molecules get changed and this creates disturbance in internal structure that pass on from one end to another end like a wave. Since force, displacement and motion of molecules are involved in this, that is why it also carries energy (potential and kinetic) and momentum along with it.



Distance between molecules increases in elastic body

Mechanical and Non-Mechanical Waves

Some waves are produced and propagated in material medium, which must possess inertia and elasticity. Such waves produced in elastic material medium are called mechanical waves or elastic waves.

There is another category of waves which requires no material medium for their propagation. They can travel even in vacuum. Such kind of waves are called non-mechanical or electromagnetic waves. Examples of such non-mechanical waves are radio waves, microwaves, light waves, X-rays and γ-rays. Electromagnetic waves are produced due to changes of electric and magnetic fields associated with moving charges.

MECHANICAL WAVES TYPES:

According to the mechanical properties of the medium, and in other ways, mechanical waves can be classified as.

1. Direction of Particle Motion:

On the basis of the direction of motion of the particles in the medium, waves can be classified as,

A **transverse wave** is formed when motion of particles is perpendicular to the direction of disturbance (direction of wave). String wave in guitar is an example of transverse wave.

If motion of particles is along the motion of disturbance (wave) then it is called **longitudinal wave**. Sound wave is an example of longitudinal wave.

2. Number of Dimensions:

Dimensions of wave is the number of independent direction in which disturbance of wave exists in space.

Wave pulse in a stretched string is **one-dimensional wave**. Wave produced on water surface of pond when stone is thrown is a **two-dimensional wave**. A sound wave and electromagnetic wave are example of **three-dimensional wave**.

3. Periodicity:

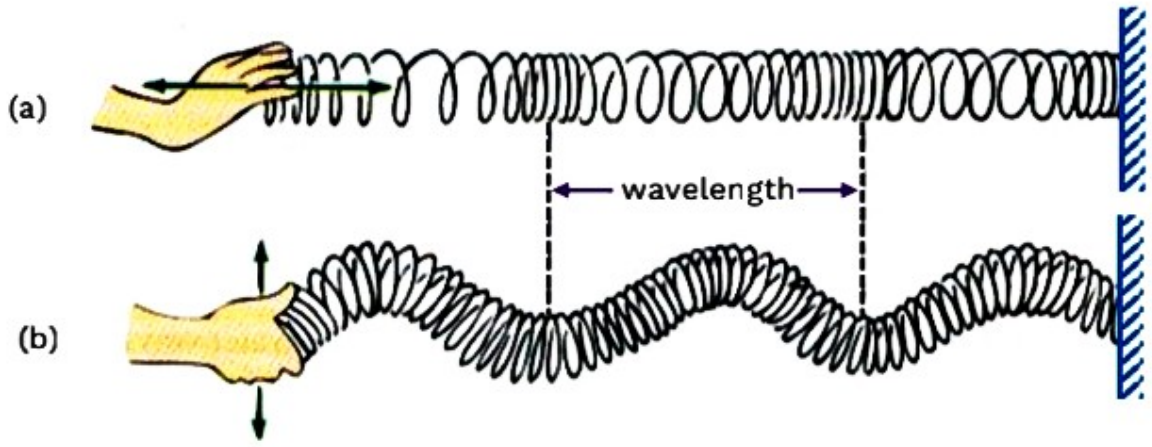
If source of a wave by which it is created is periodic then the resultant disturbance (wave) will be periodic. For example a wave pulse created when stone is dropped in a pond would be periodic if similar stones are dropped in even time intervals.

WAVE FRONTS SHAPES

The shape formed by different particles which are vibrating in similar manner (having same phase) is called its wavefront. The waves formed by the rock that falls into the lake are circular in shape have circular wave front. The sound wave propagating from a point outward has spherical wavefront. The plane wave moving in one direction in 3-D space have a plane wavefront. (The waves appear to be plane waves far away from a point source emitting spherical waves).

Solid can support Transverse as well as Longitudinal wave as it provide resistance to tensile (compression) and shear forces. The liquid does not have a defined state or structure to maintain and provide greater resistance to compression than to shear strength. As a result, only the longitudinal wave can propagate through the gas or into a ideal liquid body. However, transverse waves can exit on surface of liquid. In the case of ripples in a lake, the force that restores the system to balance is the surface tension of the water, and for the waves of the ocean, is the gravitational force.

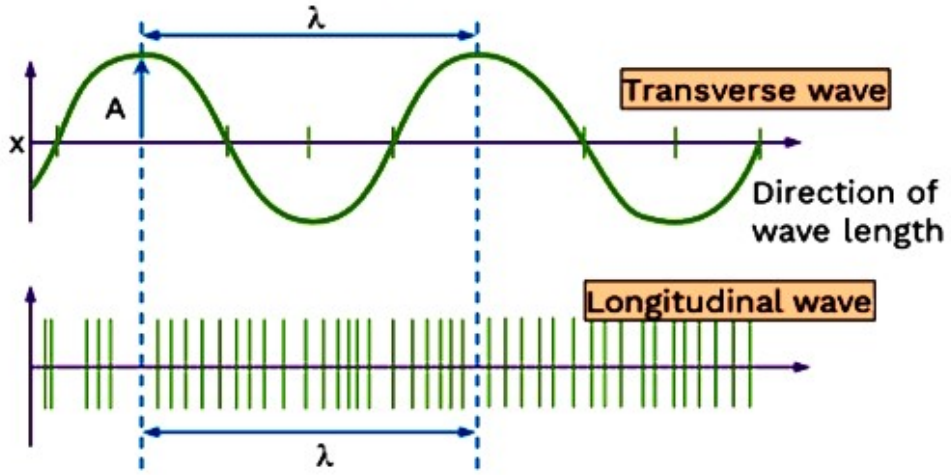
The shape of disturbance remains unchanged if there is no loss of energy during its propagation,



Describing Waves

Waves can be represented with the help of two types of graph.

(1) Snapshot showing displacement of different particles at particular instant in a transverse wave is **Displacement distance graph**.

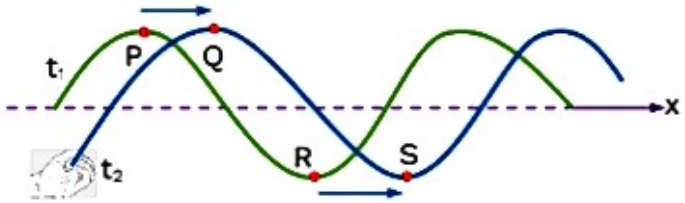


(2) **Displacement time graph** is drawn between displacement of a certain particle from mean position (undisturbed position) and time t

Wave Speed

Wave speed is the distance travelled by the disturbance created at a point of space in unit time.

If we draw displacement distance graph for a string wave.



The figure shows displacement distance waveforms at two instances of time t_1 and t_2. The disturbance peak of string wave at P has moved to Q in time $t_2 - t_1$. Then

$$v = \text{wave speed} = \frac{PQ}{t_2 - t_1} = \left(\frac{RS}{t_2 - t_1}\right)$$

Amplitude:

Amplitude is the maximum displacement of any particle from it's undisturbed position (mean position) on either side.

Wavelength:

Distance between two nearest particles which are vibrating in same phase along the direction of propagation of wave is called Wavelength λ. It can also be defined as the distance travelled by the wave in one time period of oscillation.

Time Period:
Time period T is the minimum time taken by oscillating particle to repeat the position of maximum displacement on one side.

Wave Frequency:

Wave frequency f is the number of complete cycles (oscillations) made by a particle during 1 second of its motion. It's expressed in Hertz (1 Hz = 1 cycle/s).

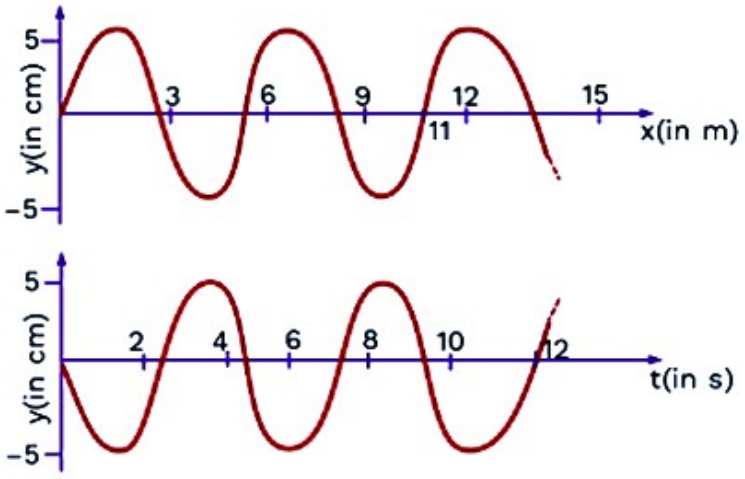
Q **A sinusoidal wave travelling along a rope completes 40.0 vibrations in 30.0 s. It also travels 425 cm along the rope in 10.0 s. Find wave speed and wavelength of this wave.**

Sol. From the definition of frequency its frequency $f = \frac{40.0\,\text{waves}}{30.0\,\text{s}} = 1.33\ \text{s}^{-1}$

And the wave speed is $v = \frac{425\,\text{cm}}{10.0\,\text{s}} = 42.5\ \text{cm/s}$

Since $v = \lambda f$, the wavelength is $\lambda = \frac{v}{f} = \frac{42.5\,\text{cm/s}}{1.33\,\text{s}^{-1}} = 0.319\ \text{m}$

Q **A transverse wave is travelling along a horizontal string. The first is displacement distance graph the second is displacement time graph showing vertical displacement of a point on the string as a function of time. Find the wave speed.**



Sol. From x-y graph, the length of two waves is 11 m.

Hence wavelength $\lambda = \frac{11}{2} = 5.5$ m

From y-t graph it is clear that a point completes 2.5 vibrations in 12 sec.

Hence time period $T = \frac{12}{2.5} = 4.8$ sec.

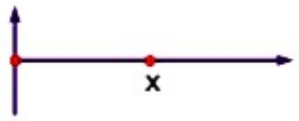
wave speed, $v = \frac{\lambda}{T} = \frac{5.5}{4.8} = 1.146$ m/s

Wave Function

Wave function a mathematical function having two independent variable position(x) and time(t) which gives displacement of different particle from its mean position.

Wave function, $y = f(x, t)$

Consider a string along x-axis, its displacement at any moment is given by



$$y(x = 0, t) = f(t)$$

Now, if wave is moving in positive x direction, the motion of particle at x = 0 is repeated by particle at any point x after time $(t - \frac{x}{v})$ where v is wave speed. Then,

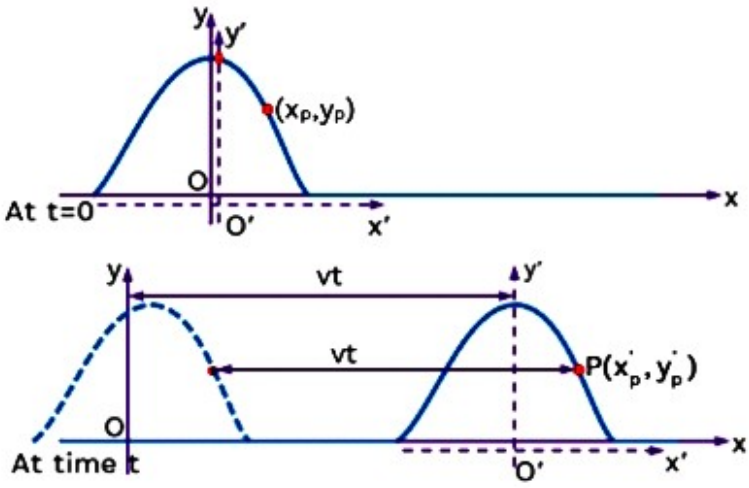
$$y(x, t) = y(x = 0, t - \frac{x}{v})$$

$$\Rightarrow y(x, t) = f(t - \frac{x}{v})$$

A wave function of wave moving in positive x direction can also be defined from function of shape of wave at t = 0, as the same shape travels on theat medium.

Let us consider shape of wave function at t = 0 is given by

$$y(x, t = 0) = f(x)$$



Displacement at particle of any general point (x_p', y_p') at any time t would be the same as by the particle at $(x_p\, y_p)$ at time t = 0.

$$y(x, t) = y(x - vt, t = 0)$$

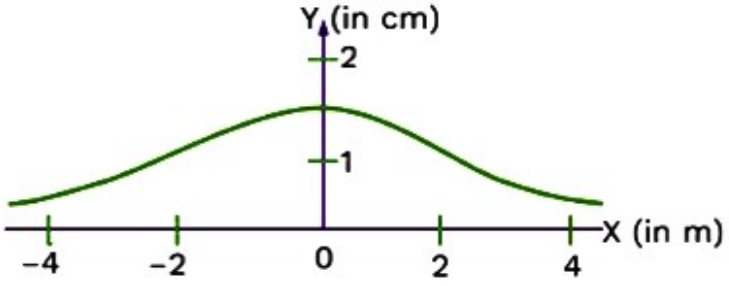
$$y(x, t) = f(x - vt).$$

Q **Wave speed of a wave pulse travelling along positive x direction on a string is 2 m/s and its displacement y (in cm) of particle at origin at any time t is given by $2/(t^2 + 1)$. Find**
(a) Expression of wave function, y = f(x, t).
(b) Draw its displacement distance graph t = 0 and t = 1 s.

Sol. (a) By replacing t by t – (x/v), we can get the desired wave function, i.e.,

$$y = \frac{2}{\left(t - \frac{x}{2}\right)^2 + 1}$$

(b) We can use wave function at a particular instant to find the shape of the wavepulse using different values of x.



at $t = 0$, $y = \dfrac{2}{\dfrac{x^2}{4}+1}$

at $x = 0$, $y = 2$ cm

$x = 2$ cm, $y = 1$ cm

$x = -2$ cm, $y = 1$ cm

$x = 4$ cm, $y = 0.4$ cm

$x = -4$ cm, $y = 0.4$ cm

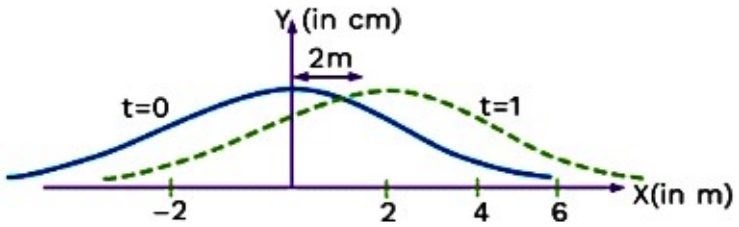
Similarly for $t = 1$ sec, displacement of particle can be calculated as follows.

at $t = 1$ s, $\quad y = \dfrac{2}{\left(1-\dfrac{x}{2}\right)^2+1}$

at $x = 2$ m, $\quad y = 2$ cm (maximum value)

at $x = 0$ $\quad y = 1$ cm

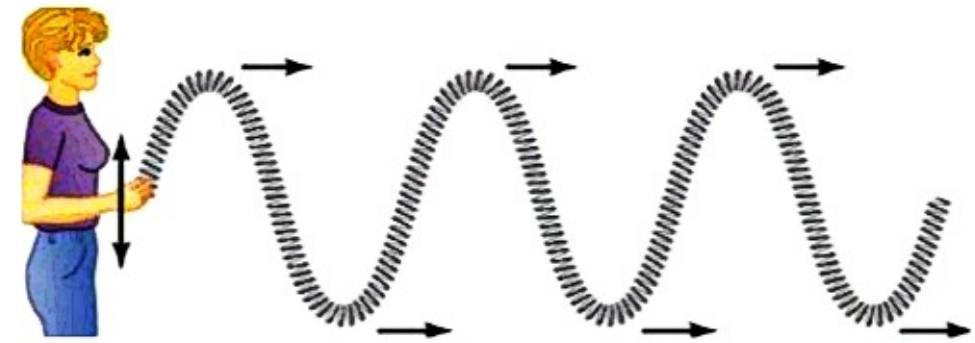
at $x = 4$ m, $\quad y = 1$ cm



The pulse has moved to the right by 2m in 1s interval.

Travelling Sine Wave in One Dimension (Wave on String)

The wave equation $y = f\left(t-\dfrac{x}{v}\right)$ is applicable for all arbitrary wave shape weather it is transverse or longitudinal. The shape of wave depends on function f. In most general cases function f is sinusoidal having shape of sine or cosine function. It is only possible when source of wave moving the left end of the string also vibrates in simple harmonic motion.



$f(t) = A \sin\omega t$

Equation of motion of particle at x = 0 may be written as

$f(t) = A \sin\omega t$

The amplitude, shape and frequency (time period) of wave depends on it source.

The particle's displacement at position x and at time t will be

$$y(x, t) = f\left(t - \frac{x}{v}\right)$$

or $$y(x, t) = A \sin \omega \left(t - \frac{x}{v}\right)$$

or $$y(x, t) = A \sin (\omega t - kx)$$

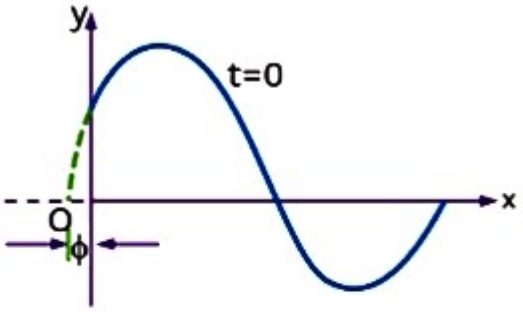
where $k = \frac{\omega}{v} = \frac{2\pi}{\lambda}$ is called wave number and $T = \frac{2\pi}{\omega} = \frac{1}{f}$ is time period of the wave, therefore most common equation can be written as

$y = A \sin (\omega t - kx + \phi)$

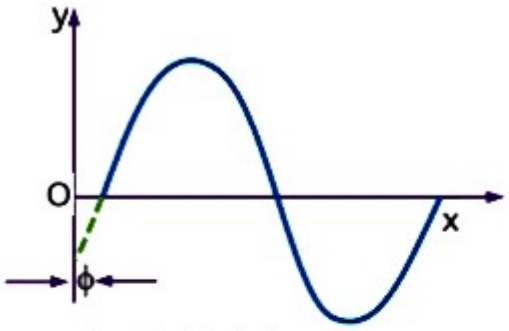
In the above equation the term $(\omega t - kx + \phi)$ is phase of the wave. ϕ is chosen to meet the initial condition. The waves with the phase differing by a multiple of 2π are known to be in phase

Positive and Negative Initial Phase Constants:

In general, the equation of harmonic wave travelling along the positive x-axis is expressed as $y = A \sin (kx - \omega t \pm \phi)$, where ϕ is called the initial phase constant. It determines the initial displacement of the particle at x = 0 when t = 0.



Positive initial phase constant
$y = A \sin (kx - \omega t + \phi)$
The sine curve starts from the Left of the origin.



Negative initial phase constant
$y = A \sin (kx - \omega t + \phi)$
The sine curve starts from the right of the origin.

Particle velocity :

It is the actual velocity of particles which are vibrating in medium. Wave equation is partially differentiated which respect to time (x is as a constant) to get the expression of particle velocity.

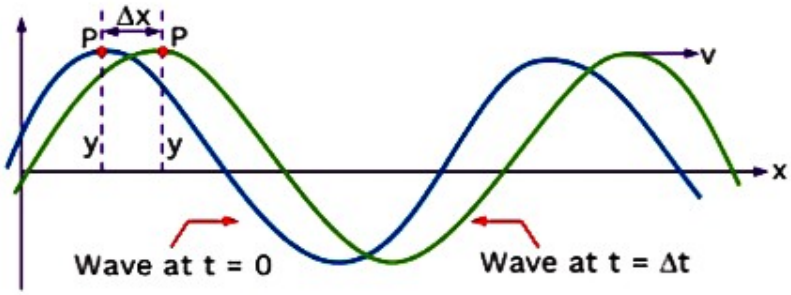
$$\frac{\partial y}{\partial t} = A\,\omega\cos(\omega t - kx + \phi)$$

Here it is to be noted that wave velocity (dx/dt) is different from particle velocity as wave velocity is a constant and depends on a medium while particle velocity changes both in magnitude and direction in a cycle.

Speed of Sinusoidal Wave:

Consider a wave travelling along positive x-direction. Figure shows two snapshots of the wave at a small interval of time Δt. Let Δx be the movement of entire wave pattern in time Δt, then wave speed is defined as

$$v = \frac{\Delta x}{\Delta t} \qquad ...(i)$$



A travelling wave in the positive x-direction is represented by

$$y(x, t) = A\sin(\omega t - kx + \phi_0) \qquad ...(ii)$$

The phase of the wave is given by

$$\phi = (\omega t - kx) + \phi_0 \qquad ...(iii)$$

As long as the shape of the wave does not change, ϕ remains constant. In such a case, as t increases, x increases in such a way that $(\omega t - kx)$ is a constant. It is to be noted that ϕ_0 (initial phase angle) remains constant.

Since $(\omega t - kx)$ = a constant, $\frac{d}{dt}(\omega t - kx) = 0$

or $$\omega - k\frac{dx}{dt} = 0 \text{ or } \frac{dx}{dt} = \frac{\omega}{k}$$

But as $\frac{dx}{dt} = v$ (speed of the travelling wave),

$$v = \frac{\omega}{k} \qquad ...(iv)$$

As $\omega = 2\pi f$ and $k = 2\pi/f$, from equation (iv), we obtain

$$v = \frac{2\pi f}{2\pi/\lambda} \text{ or } v = f\lambda \qquad ...(v)$$

Equation (iv) is a general relation which is valid for all progressive waves.

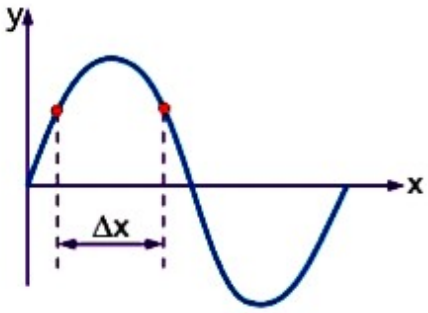
Variation of Phase With Distance

At a given instant of time t = t, phase at x= x_1,

$$[\phi]_{x_1} = 2\pi\left(\frac{t}{T} - \frac{x_1}{\lambda}\right) + \phi$$

For the wave travelling in positive x-direction and phase at x = x_2,

$$[\phi]_{x_2} = 2\pi\left(\frac{t}{T} - \frac{x_2}{\lambda}\right) + \phi$$



$$\Rightarrow \quad \Delta\phi = [\phi]_{x_1} - [\phi]_{x_2} = \frac{2\pi}{\lambda}(x_2 - x_1) = \frac{2\pi}{\lambda}\Delta x$$

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta x$$

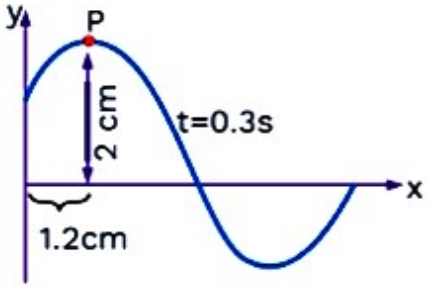
i.e., Phase difference = $\frac{2\pi}{\lambda}$ × Path difference

For $\Delta x = m\frac{\lambda}{2}$, where m = 1, 3, 5, 7, ...

$\Delta\phi = m \times \pi$

Points separated by distance m(λ/2), where m is an odd integer are out of phase (at any instant of time).

Q **The wavelength and amplitude of a sinusoidal travelling wave are 7.5 cm and 20 cm respectively. Write the equation of travelling wave whose displacement-distance graph is given at t = 0.3 s (if the crest P was at x = 0 at t = 0).**



Sol. Given, A = 2 cm, λ = 7.5 cm

$$K = \frac{2\pi}{\lambda} = \frac{4\pi}{15}\text{cm}^{-1}$$

In 0.3 sec wave have has travelled a distance of 1.2 cm then,

$v = 1.2/0.3 = 4$ cm/s

Therefore, angular frequency $\omega = (v)(k) = \frac{16\pi}{15}$ rad/s

Since crest (maximum displacement) is at $x = 0$ at $t = 0$, we can write the wave equation as,

$$y(x, t) = 2\cos(kx - \omega t) = 2\cos\left(\frac{4\pi}{15}x - \frac{16\pi}{15}t\right)$$

Differential Equation for Travelling Wave

Consider a wave function travelling towards positive x-axis.

$$y(x, t) = f\left(t - \frac{x}{v}\right)$$

Velocity of particle, $\frac{\partial y}{\partial t} = f'\left(t - \frac{x}{v}\right)[1 - 0]$ ...(i)

Acceleration of particle, $\frac{\partial^2 y}{\partial t^2} = f''\left(t - \frac{x}{v}\right).1$ ...(ii)

Slope of string, $\frac{\partial y}{\partial x} = f'\left(t - \frac{x}{v}\right)\left[\frac{-1}{v}\right]$ ...(iii)

Concavity of string, $\frac{\partial^2 y}{\partial x^2} = f''\left(t - \frac{x}{v}\right)\left[\frac{1}{v^2}\right]$ ...(iv)

From equation (ii) and equation (iv)

$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$ (This is a characteristic equation of wave)

and from equation (i) and equation (ii)

$\frac{\partial y}{\partial t} = -v.\frac{\partial y}{\partial x}$

Particle velocity = – wave velocity × slope of string.

Q **A wave equation is given by,**

$$y = 4\sin\frac{\pi}{2}\left(2t + \frac{x}{8}\right)$$

where x is in centimetres and time in seconds. Calculate

(a) the amplitude, wavelength,

(b) the phase difference of same particle for two different positions in time interval of 0.4 sec

(c) the phase difference at any given instant of two particles 12 cm apart.

Sol. The equation of a wave motion is given by

$$y = A \sin \frac{2\pi}{\lambda}(vt + x) \quad \text{...(i)}$$

Here, $y = 4 \sin \frac{\pi}{2}\left(2t + \frac{x}{8}\right)$

This equation can be written as

$$y = 4 \sin \frac{2\pi}{32}(16t + x) \quad \text{...(ii)}$$

Comparing equation (i) with equation (ii), we get,
Amplitude A = 4 cm; wavelength λ = 32 cm; wave velocity v = 16 cm/s

Here frequency is given as $f = \frac{v}{\lambda} = \frac{16}{32} = \frac{1}{2}$ = 0.5 Hz.

(a) Phase of a particle at instant t_1 is given by

$$\phi_1 = \frac{\pi}{2}\left(2t_1 + \frac{x}{8}\right)$$

The phase at instant t_2 is given by

$$\phi_2 = \frac{\pi}{2}\left(2t_2 + \frac{x}{8}\right)$$

The phase difference is given as

$$\phi_1 - \phi_2 = \frac{\pi}{2}\left[\left(2t_1 + \frac{x}{8}\right) - \left(2t_2 + \frac{x}{8}\right)\right]$$

$= \pi(t_1 - t_2) = \pi(0.4)$ (As $t_1 - t_2 = 0.4$)

$= 180 \times 0.4 = 72°$ (π rad = 180°)

(b) Phase difference at an instant between two particles with path difference Δx is

$$\phi = \left(\frac{2\pi}{\lambda} \cdot \Delta x\right) = \frac{2\pi}{32} \times 12 \qquad \text{(As } \Delta x = 12 \text{ cm)}$$

$$= \frac{3\pi}{4}$$

Q **A transverse wave travelling on a string has maximum particle velocity 3 m/s maximum acceleration 90 m/s². Write wave function describing the wave having speed 20 m/s along the string.**

Sol. Maximum particle velocity, $u_{max} = \omega A$ ...(i)

Maximum particle acceleration, $a_{max} = \omega^2 A$ ...(ii)

$\therefore$ Angular frequency, $\omega = \frac{a_{max}}{u_{max}} = \frac{90}{3} = 30$ rad/s

From equation (i), amplitude, $A = \frac{u_{max}}{\omega} = \frac{3}{30} = 0.1$ m

Propagation constant, $k = \frac{\omega}{v} = \frac{30}{20} = 1.5\ m^{-1}$

Equation of wave is $y = A \sin(\omega t \pm kx)$ or wave function $y = 0.1 \sin(30t \pm 1.5x)$.

Wave Travelling on String

Assumption:

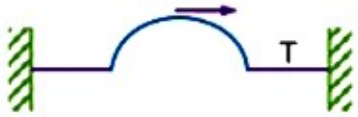
(1) String is homogeneous.

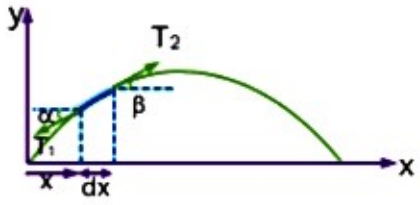
(2) It is perfectly elastic.

(3) Tension in the string is so large that we can neglect effect of gravity.

(4) Particles on string strictly perform small transverse oscillations.

(5) Slope of string at every point is very small.





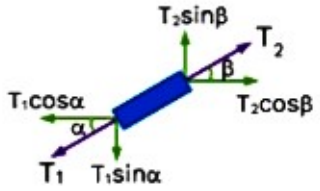
Now drawing free body diagram of small element

Since, there is no motion in x direction,

$\sum F_x = 0$

$\Rightarrow T_2\cos\beta - T_1\cos\alpha = 0$

$\Rightarrow T_1\cos\alpha = T_2\cos\beta = T$ ...(i)



and in y direction

$\sum F_y = ma$

$\Rightarrow T_2\sin\beta - T_1\sin\alpha = dm(a)$

Using value of T_1 and T_2 from equation (i) we get,

$\Rightarrow \frac{T}{\cos\beta}\sin\beta - \frac{T}{\cos\alpha}\sin\alpha = \mu dx\frac{\partial^2 y}{\partial t^2}$, μ = Linear mass density of string.

$\Rightarrow T\left(\tan\beta - \tan\alpha\right) = \mu dx\frac{\partial^2 y}{\partial t^2}$

$\Rightarrow \lim_{\partial x\to 0}\frac{\left(\frac{\partial y}{\partial x}\right)_{x+\partial x} - \left(\frac{\partial y}{\partial x}\right)_x}{\partial x} = \frac{\mu}{T}\frac{\partial^2 y}{\partial t^2}$

$\Rightarrow \frac{\partial^2 y}{\partial x^2} = \frac{\mu}{T}\frac{\partial^2 y}{\partial t^2}$

On comparing with characteristic equation of wave that is

$\Rightarrow \boxed{v^2\frac{\partial^2 y}{\partial x^2} = \frac{\partial^2 y}{\partial^2 t}}$

$\Rightarrow \frac{1}{v^2} = \frac{\mu}{T}$

$\Rightarrow v = \sqrt{\frac{T}{\mu}}$

$\Rightarrow v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{\rho A}} = \sqrt{\frac{\text{stress}}{\rho}}$

$\Rightarrow v = \sqrt{\frac{y\times\text{strain}}{\rho}}$ → Mechanical wave requires inertia (ρ) and elasticity (stress) to travel in a medium.

Transverse Tension:

$$T_x = T = T_1\cos\alpha$$

$$T_y = T_1\sin\alpha$$

$$\tan\alpha = \frac{T_y}{T_1} \Rightarrow \boxed{T_y = T_x\frac{\partial y}{\partial x}}$$

Q **Find speed of the wave generated in the string having linear mass density 500 gm/m. (Tension in the chord is not effected by its mass).**



Sol. $T = 20 \times 10 = 200N$

$$v = \sqrt{\frac{200}{0.5}} = 20 \text{ m/s}$$

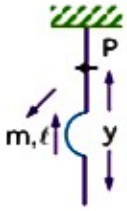
Q **A uniform rope having mass m and length L hangs from ceiling as shown figure,**
(a) Find the speed of a transverse wave as a function y.
(b) Find the time taken by wave pulse to travel entire length of the rope.

Sol. (a) As mas per unit length

$$\mu = \frac{m}{\ell}$$

$$\therefore \quad V = \sqrt{\frac{T}{\mu}}$$

$\therefore$ Tension at P = μyg



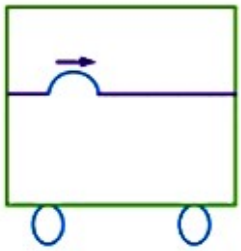
$$\therefore \quad V = \sqrt{\frac{\mu yg}{\mu}} = \sqrt{yg}$$

(b) Now $\frac{dy}{dt} = \sqrt{yg}$

$$\int_0^\ell \frac{dy}{\sqrt{y}} = \sqrt{g}\int_0^t dt$$

$$t = 2\sqrt{\ell / g}$$

Q **A stretched string is used to generate a wave pulse staring at the left end is given in figure. Find the velocity of the cart for the pulse to remains stationary w.r.t ground if tension in string is 100 N and linear mass density is 0.25 kg/m.**



Sol. Velocity of pulse = $\sqrt{\frac{T}{\mu}}$ = 20 m/s

Now $\vec{v}_{PG} = \vec{v}_{PC} + \vec{v}_{CG}$

$0 = 20\hat{i} + \vec{v}_{CG}$

$\vec{v}_{CG} = -20\hat{i}$ m/s

TRANSMITTED POWER ALONG THE STRING THROUGH A SINE WAVE

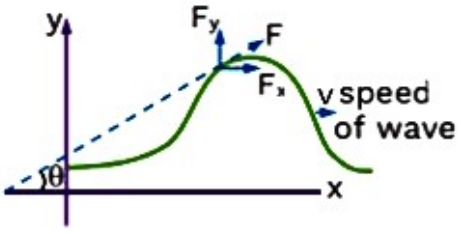
When a travelling wave is on a string, energy gets transmitted in direction of propagating wave, in two different forms : kinetic energy and potential energy.

Average Power (P) can be expressed as, $P_{avg.} = 2\pi^2 f^2 A^2 \mu v$.

Derivation of Average Power:

Suppose equation of the wave is:

$$y = A \sin\left(\omega\left(t - \frac{x}{v}\right)\right)$$



Power = $F_y V_y$

$$P = (F\sin\theta)\frac{\delta y}{\delta t} = F\sin\theta \frac{\delta y}{\delta x} \times \frac{\delta x}{\delta t}$$

Here F is tension in the string $F = \mu v^2$ and

For small angle $\sin\theta = \tan\theta = \frac{\delta y}{\delta x}$ and as $y = A \sin\left(\omega\left(t - \frac{x}{v}\right)\right)$ so

$$\frac{\delta y}{\delta x} = -\frac{A\omega}{v}\cos\left(\omega\left(t - \frac{x}{v}\right)\right) \text{ and}$$

$$\frac{\delta y}{\delta t} = A\omega\cos\left(\omega\left(t - \frac{x}{v}\right)\right)$$

Putting the values $P = (\mu v^2)\left(\frac{\delta y}{\delta x}\right)\frac{\delta y}{\delta x}\frac{\delta x}{\delta t}$

$$\frac{\delta x}{\delta t} = v \text{ so, } P = \mu v^3\left(\frac{\delta y}{\delta x}\right)^2$$

$$P = \mu v^3\left(-\frac{A\omega}{v}\cos\left(\omega\left(t - \frac{x}{v}\right)\right)\right)^2$$

or $P = \mu v A^2\omega^2 \cos^2\left(\omega\left(t - \frac{x}{v}\right)\right)$

$\therefore$ Average value of $\cos^2\left(\omega\left(t - \frac{x}{v}\right)\right) = \frac{1}{2}$ so

Average power $\langle P \rangle = \frac{1}{2}\mu v A^2\omega^2$

As $\omega = 2\pi f$

So Average Power $\langle P \rangle = 2\pi^2 f^2 A^2 \mu v$

Energy Transferred = $\int_0^T P_{av}\,dt$

Energy transferred in one time = PavT(energy stored in one wavelength)

Intensity: It is amount of energy transferred per second per unit cross section area perpendicular to the direction of travelling wave.

$$I = \frac{\text{Power}}{\text{Cross sectional area}} = \frac{P}{s} \Rightarrow I = \frac{1}{2}\rho\omega^2 A^2 v$$

The above expression gives average intensity of a wave.

Energy density is energy per unit volume and it is given by.

$$= \frac{Pdt}{svdt} = \frac{I}{v} = \frac{1}{2}\rho\omega^2 A^2$$

Q **A string under a tension of 80.0 N having linear mass density 5.0×10^{-2} kg/m generates a sinusoidal wave at a frequency of 60 Hz at amplitude of 6 cm. Calculate the amount of power supplied to the string?**

Sol. Wave speed, $v = \sqrt{\frac{T}{\mu}} = \left(\frac{80.0\,N}{5.00 \times 10^{-2}\,kg/m}\right)^{1/2} = 40$ m/s

Angular frequency $\omega = 2\pi f = 2\pi(60.0\ Hz) = 377\ s^{-1}$

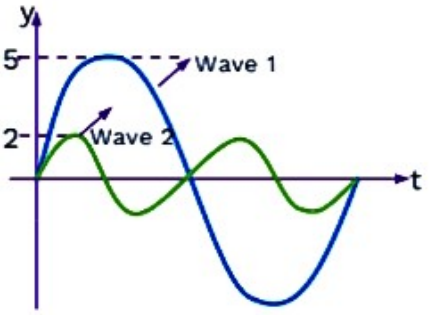
Power supplied is given by,

$$p = \frac{1}{2}\mu\omega^2 A^2 v$$

$$= \frac{1}{2}(5.0 \times 10^{-2}\ kg/m)\ (377\ s^{-1})^2 \times (6.00 \times 10^{-2}\ m)^2\ (40\ m/s)$$

$$= 512\ W.$$

Q **Displacement distance curve of two waves in the same medium are represented by y-t curves in the figure. Find ratio of their average intensities?**



Sol. $\frac{I_1}{I_2} = \frac{\omega_1^2 A_1^2}{\omega_2^2 A_2^2} = \frac{f_1^2 A_1^2}{f_2^2 A_2^2} = \frac{1 \times 25}{4 \times 4} = \frac{25}{16}$